

TM Forum Component

Service Test Management

TMFC055

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Notice

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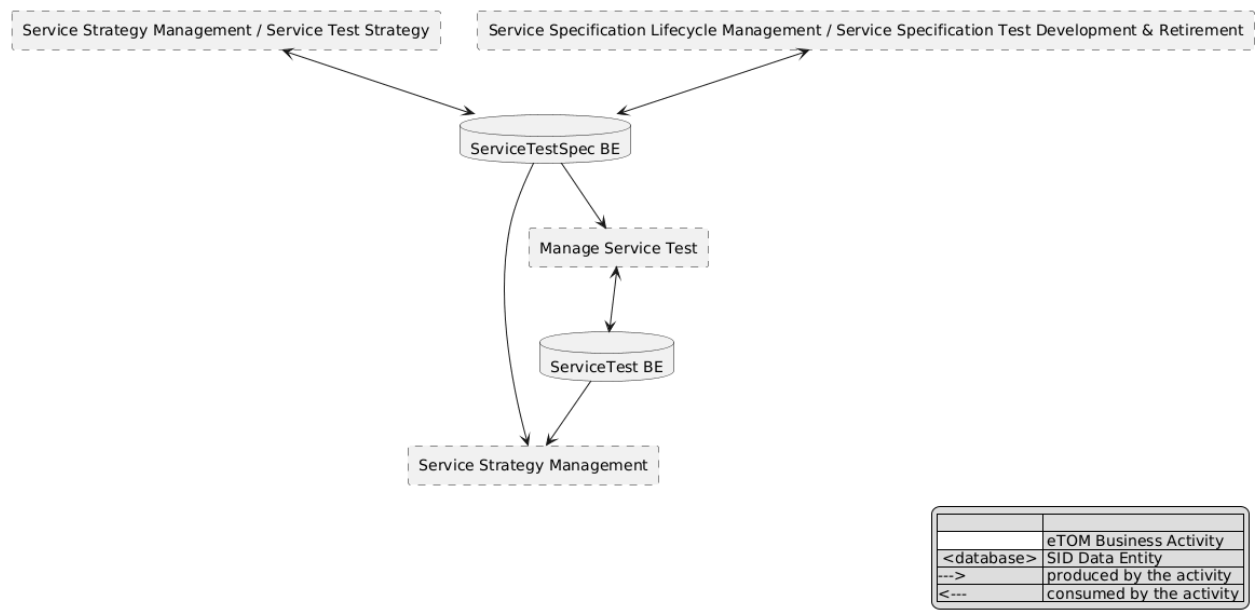
TMFC055 – Service Test Management

1. Overview

Component Name	ID	Description	ODA Function Block
Service Test Management	TMFC055	Service Test Management is responsible for managing the service test specification and executing tests based on this specification to ensure the service operates correctly. Service test specification involves capturing potential tests for each Service Specification that can be initiated either automatically or manually by the service user. These tests are derived from those available at the service specification or resource specification level. The scope of the component includes the end-to-end execution and reporting of tests, tailored to the specific context and actor involved.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.4.1	L2	Service Strategy Management	Enable the development of a strategic view and a multi-year business plan for the enterprise's services and service directions, and the parties who will supply the required services.

1.4.1.8	L3	Service Test Strategy	Service Test Strategy develops the strategies of the enterprise for Service Test. / This process is in charge of identifying types of Service Test to be conducted according to different context (i.e. business activities) for types of Services. / The different context for Service Test are: / - Service Development & Retirement to qualify the capacity to deliver Services before validating a new ServiceSpecification / - Service Configuration & Activation to test the Service before closing the ServiceOrderItem / - Service Quality Management to test Service Quality / - Service Problem Management to test Service functioning / - Service Test Management to conduct tests not specific to a customer's product
1.4.3.8	L3	Service Specification Test Development & Retirement	Service Test Development & Retirement is in charge of the Service Test catalogue. / A type of Service Test aims at measuring proper functioning and capacities of a Service. / Service Test Development & Retirement includes: / - Specifying in detail each Service Test according to the different context. It includes specifying: / - the roles authorized to use the Test and quotas for each type of role / - the method to conduct the Test / - the rules that define the strategies for conducting the test (including the test plan) / - the thresholds and related actions / - the relationships with lower level tests (Resource Test) / - the report of test results with rules for enrichment of Resource Tests results according to role asking for it / - Specifying test scenarios defining sequence of Tests with rules about context and planning to trigger it. It includes roles allowed for asking test scenario and corresponding quotas.

1.4.1.9	L3	Analyze Service Test Quality	Analyze Service Test Quality process performs quality analysis of service testing processes, in order to continuously fine-tune and improve them. / Service Test Quality Analysis processes provide an offline analysis of the performance of service testing processes to ensure their continuous fitness for purpose, and eventually propose modifications or enhancements to the current service test specifications (including: service testing methods, quota, authorized users, recommendations and course of action if needed). / The analysis is done using statistical service test usage data to evaluate service tests performance and required improvements if needed. / Following the analysis, typical course of action can be: / - No changes needed currently, / - Creation of additional service tests, / - Improving existing service tests, / - Removing service tests that are no longer relevant.
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1.4.4.6	L3	Manage Service Test	Manage Service Test process manages the end-to-end execution of a test or test scenario for services not specific to a customer. Tests can be manual or automated. / Service Test Management processes rely on Product and Resource Test Management processes due to dependencies between product tests and resource tests. / Service Test Management includes: - Identification of service tests needed according to fulfillment and assurance issues - triggering of service tests in manual or automated mode - Execution of service tests - Verification of test authorization (role and context) and quota management - Identification and prioritization of tests - Setting up the test context and configuration - Triggering appropriate product and resource Tests - Tests can be on-demand or planned according to specific needs - Enrichment with product and resource tests results based on applicable service test rules - Reporting of test results
1.4.4	L2	Service Support Management	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes.

1.4.3	L2	Service Specification Lifecycle Management	Service Specification Lifecycle Management processes are project oriented in that they develop and deliver new or enhanced service types. These processes include process and procedure implementation; systems changes and customer documentation. They also undertake rollout and testing of the service type, capacity management and costing of the service type. It ensures the ability of the enterprise to deliver service types according to requirements. / / This process was renamed in 24.0. Old name was- Service Specification Development & Retirement.
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2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Service Test	The Service Test ABE defines measurement including units and values and how the values are determined. It also includes thresholds used to evaluate the measure and the consequences of violating the thresholds as well as the Service Test measures that reflect the functioning of the Service under test.	ServiceTest	An entity that exists for a controlled test invocation on a service. The service test is executed according to a schedule. and contains service test configuration parameters that are to be applied at execution time, and service test measures that result.

Service Test	The Service Test ABE defines measurement including units and values and how the values are determined. It also includes thresholds used to evaluate the measure and the consequences of violating the thresholds as well as the Service Test measures that reflect the functioning of the Service under test.	ServiceTestSpec	The service test specification describes the service test in terms of parameters to be configured and measures to be taken.
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2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
625	Service Test Operation Control	Service Capability Management	Service Test Resource Check	Service Test Operation Control function provides the necessary functionality to access, command and control the various service test devices required to perform service testing.

1136	Service Test Specification and Scenario Management	Service Test Specification Development	Service Test Policy Management	Service Test Specification and Scenario Management Function includes: / / Test scenario management: for example, a home equipment self-test can trigger a test scenario that verifies the functioning of all the services associated, depending on the installed products of the user. Thus, the box could test its internal configuration and functioning and its capacity to connect to internet and VoIP services. / Managing service test capabilities: some resources used to implement services have limited capabilities for executing tests, especially for multiple parallel testing. All the constraints linked to limited resources test capabilities and depending on the execution context must be aggregated and managed at service test level. / Managing initiator authorization: some tests are restricted to some user roles (internal or external) so the authorization are defined at the Service Test Strategy and Policy Management.
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566	Service Test Result Analysis Policy Configuration	Service Test Specification Development	Service Test Policy Management	Service Test Result Analysis Policy Configuration functions provide the necessary functionality to manage how the test results should be interpreted. / Strategies can range from simple (e.g. - check the status of a router) to complex (e.g. - perform an end-to-end test on a circuit and sectionalize the problem).
573	Service Testing Rules Management	Service Test Specification Development	Service Test Policy Management	Service Testing Rules Management provides management of service testing rules.
581	Automated Service Test Invocation	Service Test Management	Service Test Conducting	Automated Service Test Invocation provides automated invocation of Service Test Services for test and retrieval of results from the resource testing capabilities.

1138	Service Test Reporting Rules Management	Service Test Specification Development	Service Test Policy Management	Service Test Reporting Rules Management Function includes: // Test rules defining the enrichment of test results. // - For example: enrichment of internet bandwidth measure based on installed product // Test rules defining the restitution of the result depending on test execution context // - For example: a test result of ok/not ok can be enough for a user whereas customer care requires more details in the result of the test.
582	Manual Service Test Invocation	Service Test Management	Service Test Conducting	Manual Service Test Invocation provides a user interface (GUI) function that provides the means to access the testing capabilities.
627	Service Test Strategy Management	Service Strategy Management	Service Strategy Design and Planning	Service Test Strategy Management functions provide the necessary functionality to manage the local rules that define the strategies for conducting a test as well as how the test results should be interpreted.

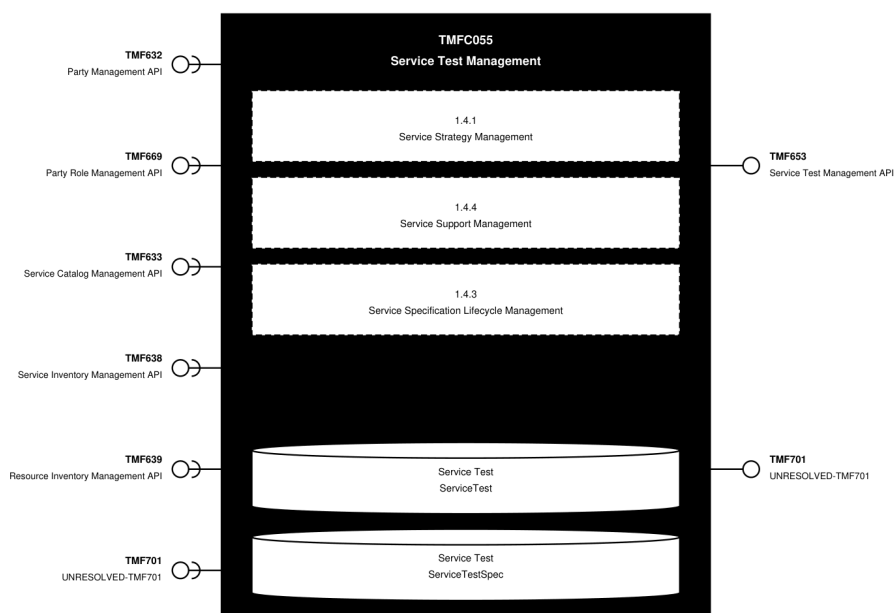
1137	Service Test Rules Management	Service Test Specification Development	Service Test Policy Management	Service Test Rules Management Function includes: // Test rules defining the strategies for setting up test configuration / Test rules defining how to carry out the service test. // - For example: to choose the next test in a scenario depending on the results of the previous tests.
575	Service Test Results Reporting	Service Test Management	Service Test Reporting	Service Test Results Reporting provides reporting of service end-to-end test results back to the client.
626	Service Testing Management	Service Test Management	Service Test Repository Management	Service Testing Management functions provides the necessary functionality to manage the end-to-end lifecycle of a test of a service. / Including the management of scheduling, retrieval of appropriate inventory data, setting up the test configuration, acquisition and management of test resources, test results interpretation, reporting of test results and close down of the test of the service.

577	Service End to End Test Scheduling	Service Test Specification Development	Service Test Policy Management	Service End to End Test Scheduling provides scheduling of end-to-end testing of services.
579	Service End to End Testing	Service Test Management	Service Test Conducting	Service End to End Testing provides test execution of the end-to-end service testing.

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

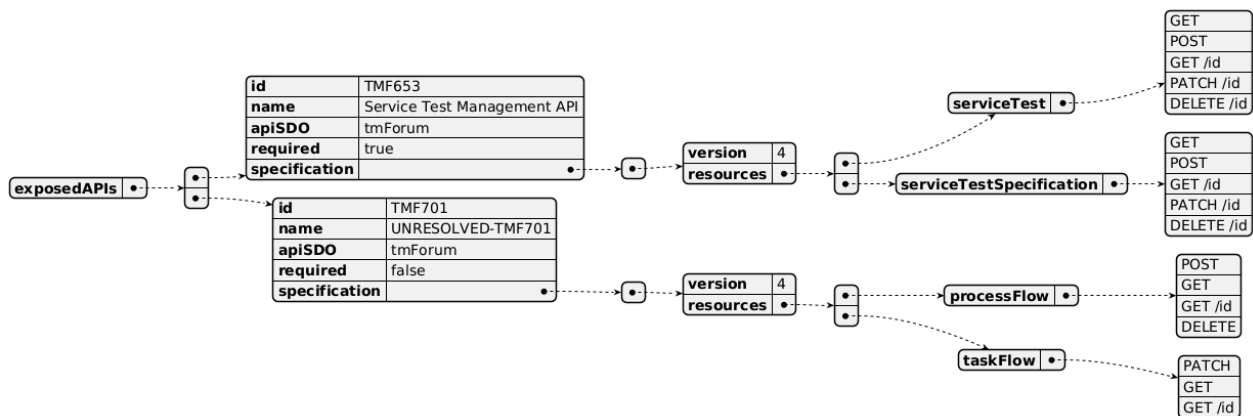
3.1. API Context Diagram



(SVG source: [TMFC055 API Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

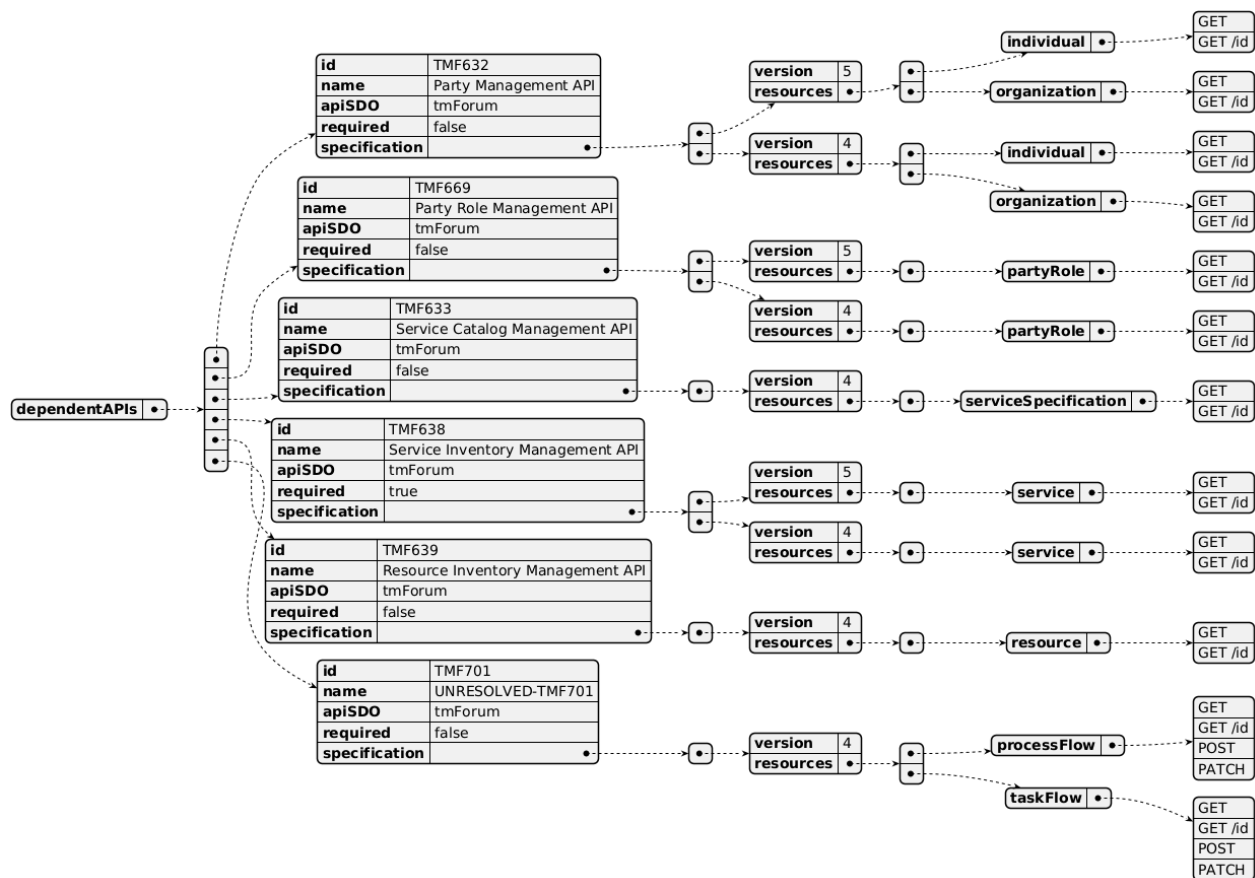
API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF653	Service Test Management API	Mandatory	4	serviceTest	GET, POST, GET /id, PATCH /id, DELETE /id
TMF653	Service Test Management API	Mandatory	4	serviceTestSpecification	GET, POST, GET /id, PATCH /id, DELETE /id
TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	PATCH, GET, GET /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id

TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id
TMF669	Party Role Management API	5	Optional	partyRole	GET, GET /id
TMF669	Party Role Management API	4	Optional	partyRole	GET, GET /id
TMF633	Service Catalog Management API	4	Optional	serviceSpecification	GET, GET /id
TMF638	Service Inventory Management API	5	Mandatory	service	GET, GET /id
TMF638	Service Inventory Management API	4	Mandatory	service	GET, GET /id
TMF639	Resource Inventory Management API	4	Optional	resource	GET, GET /id
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	GET, GET /id, POST, PATCH
TMF701	UNRESOLVED-TMF701	4	Optional	taskFlow	GET, GET /id, POST, PATCH



3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

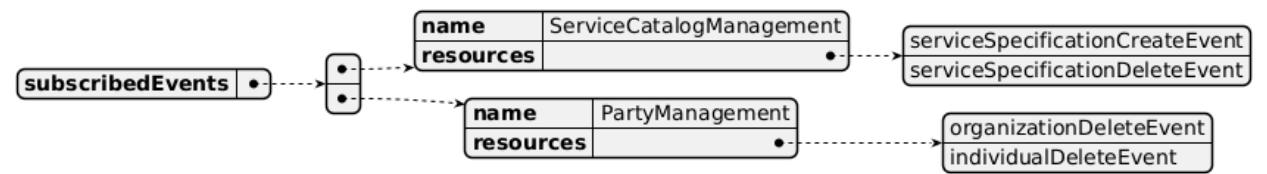
Published Events

API ID	API Name	Event Resources
TMF653	Service Test Management API	serviceTestCreateEvent serviceTestStateChangeEvent serviceTestSpecificationCreateEvent serviceTestSpecificationDeleteEvent serviceTestAttributeValueChangeEvent serviceTestDeleteEvent serviceTestSpecificationAttributeValueChangeEvent



Subscribed Events

API ID	API Name	Event Resources
TMF633	Service Catalog Management API	serviceSpecificationCreateEvent serviceSpecificationDeleteEvent
TMF632	Party Management API	organizationDeleteEvent individualDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v24.5
SID	v25.0
Functional Framework	v24.0

5.2. Jira References

Note: Same needs as at Service level for non-tangible products, plus addition of Resource Inventory APIs, to be consistent.

SID [ISA-1318] Verify role of Test Measure, Test Measure Definition and Test Specification in ServiceTestABE - TM Forum JIRA

Note: Jira ISA-1318 is raised for an open point - whether Test Measure, Test Measure Definition and Test Specification Role should be included in ServiceTestABE under Service, Resource, Product tabs.

[ISA-1251] Improve the description of Product/Service/Resource Test ABEs - TM Forum JIRA

The links between ServiceTest and ResourceTest need to be described, at specification and instance levels. For example: - a ServiceTestSpecification related to a Service should correspond to an existing ResourceTestSpecification - a ServiceTest related to a Service should trigger a ResourceTest on the corresponding Resource.

APIs As: - a ServiceSpecification related to a Service should correspond to an existing ResourceTestSpecification; - a ServiceTest related to a Service should trigger a ResourceTest on the corresponding Resource.

[AP-6690] TMF769 - add reference to ServiceTest and ResourceTest - TM Forum JIRA: TMF769 Product Test API is only available in preview, but contains no links to Service Test or Resource Test, where the real tests are done.

[AP-6691] New API: Resource Test API - TM Forum JIRA: no Resource Test Management API identified yet.

5.3. Further resources

1. IG1242: please refer to IG1242 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
Draft version 01	02-May-2025	Ritu Arora	Initial draft created

Draft Version 0.2	13-May-2025	Ritu Arora	Based on Review on 06th May: SID ABE table containing pattern ABE (TestMeasure, TestMeasureDefinition, TestSpecificationRole, TestSpecification, Test) has been removed and a Jira raised to request clarification for Test Measure, Test Measure Definition and Test Specification Role to be included in ServiceTestABE under Service, Resource, Product tabs. Under Functional Framework: Service Test Result Analysis Policy Configuration continues to be included; Service Test Resources Availability Management has been removed. Dependent APIs: Incident and Trouble Ticket Management APIs have been removed (ServiceOrderManagement API was suggested to be added as a dependent API but later suggested otherwise, so has not been included). All diagrams updated to incorporate these changes.
Draft Version 0.3	30-May-2025	Ritu Arora	Based on Review on 13th May: eTOM entries updated to include Service Specification Test Development & Retirement; Dependent APIs Product Test and Product Inventory have been removed. All diagrams updated to incorporate these changes.
Draft Version 0.4	23-Jun-2025	Ritu Arora	Changed Service Catalog API to optional in the dependent API list; respective diagram updated to reflect this change
1.0.0	18-Jul-2025	Rosie Wilson	Final Administrative Updates
1.1.0	25-May-2026	Hugo Vaughan	Gen 5 API updates
1.1.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	18-Jul-2025	Rosie Wilson	Initial release
Production	18-Aug-2025	Adrienne Walcott	Updated to Member Evaluated status

Pre-production	25-Jun-2026	Sannah Sibaya	New Release Version 1.1.0
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6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team.

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